

SECTION A: (20 MARKS)
MULTIPLE CHOICE QUESTIONS
ANSWER ALL QUESTION

1. Which of the following is wrong about constructors?
 - a. A constructor cannot be inherited by subclass.
 - b. Constructors have return types.
 - c. A class can have more than one constructor.
 - d. A constructor's name must be the same as the class name.
2. What is the process of defining more than one method in a class?
 - a. Method overriding
 - b. Method overloading
 - c. Method invocation
 - d. Method declaration
3. What is the purpose of encapsulation?
 - a. to hide the implementation details of a class and expose only the necessary features
 - b. for declaring variables in a class
 - c. to define methods in a class
 - d. to achieve multiple inheritance
4. Which feature of OOPS derives the class from another class?
 - a. Data hiding
 - b. Encapsulation
 - c. Inheritance
 - d. Polymorphism
5. Which one of the following is not applicable to an Actor in a Use Case Diagram?
 - a. An actor can interact with any number of use cases.
 - b. An actor can interact directly with other actors in a use case diagram.
 - c. An actor must be placed outside the system boundary.
 - d. An actor specifies a role played by a user or an external system that interacts with the system being depicted.
6. Which option is false about the *final* keyword?
 - a. A *final* method cannot be overridden in its subclasses.
 - b. A *final* class cannot be extended.
 - c. A *final* class cannot extend other classes.
 - d. A *final* method can be inherited.
7. A PC is composed of a motherboard, disk drives, and I/O devices. The PC is sold as a single unit without optional configurations. In a class diagram, this example should be modelled as

- a. Association
- b. Aggregation
- c. Composition
- d. Generalisation

8. What will be the output of the following program?

```
class Car{
    public Car()
    {
        System.out.println("Class Car");
    }
    public void vehicleType()
    {
        System.out.println("Vehicle Type: Car");
    }
}
class Toyota extends Car{
    public Toyota()
    {
        System.out.println("Class Toyota");
    }

    public void speed()
    {
        System.out.println("Max: 90Kmph");
    }
}
public class Vios extends Toyota{

    public Vios()
    {
        System.out.println("Toyota Model: Vios");
    }
    public void speed()
    {
        System.out.println("Max: 80Kmph");
    }
    public static void main(String args[])
    {
        Vios myCar = new Vios();
        myCar.vehicleType();
        myCar.speed();
    }
}
```

a.

```

Class Toyota
Toyota Model: Vios
Vehicle Type: Car
Max: 90Kmph

```

c.

```

Class Car
Class Toyota
Toyota Model: Vios
Vehicle Type: Car
Max: 90Kmph

```

b.

```

Class Car
Class Toyota
Toyota Model: Vios
Vehicle Type: Car
Max: 80Kmph

```

d.

```

Class Car
Class Toyota
Vehicle Type: Car
Max: 90Kmph

```

9. What will be the output of the following program?

```

interface calculate
{
    void cal(int item);
}

class display implements calculate
{
    int x;
    public void cal(int item)
    {
        x = item * item;
    }
}

class interfaces
{
    public static void main(String args[])
    {
        display arr = new display();
        arr.x = 0;
        arr.cal(-2);
        System.out.print(arr.x);
    }
}

```

- a. 0
- b. -2
- c. 4
- d. -4

10. How are *associations* modelled in a class diagram?
- Through the objects of classes
 - Through constructor
 - Through methods of classes
 - Through inheritance
11. Which of the following component gives a drop-down list of options from which to choose?
- JList
 - JButton
 - JComboBox
 - JDropDownList
12. What is the purpose of the “catch” block in exception handling?
- It specifies the exception message.
 - It specifies the exception type.
 - It contains the code that might throw an exception
 - It catches exceptions and handles them.
13. What is the output of the following code snippet?

```
try {  
    int [] arr = new int[5];  
    int value = arr[0];  
  
}  
catch (NullPointerException e){  
    System.out.println("NullPointerException!");  
}  
finally {  
    System.out.println("Finally Block!");  
}
```

- The program hangs
 - NullPointerException!
 - Compilation error
 - Finally Block!
14. Which class in Java is used to read data line by line from a file?
- BufferedReader
 - FileInputStream
 - FileWriter
 - OutputStream

15. Study the following code snippet carefully.

```
File file = new File("write.txt");
FileWriter fr = new FileWriter(file, true);
fr.write("data");
fr.close();
```

What does the flag “true” means in the following statement from the code snippet above?

```
FileWriter fr = new FileWriter(file, true);
```

- a. Appending text data to a text file
- b. Appending binary data to a binary file
- c. Appending buffered data to a text or a binary file
- d. Checking if the file exists

16. What is the output of the following program if the input given is “B@nan@s”.

```
class TestBufferedReader {
    public static void main(String[] args) {
        try {
            char c;
            BufferedReader br = new BufferedReader(new
                InputStreamReader(System.in));
            do {
                c = (char) br.read();
                System.out.print(c);
            } while (c != '@');
        } catch (IOException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

- a. B@nan@s
- b. B@nan@
- c. B@nan
- d. B@

17. In an abstract class, what does an abstract method represent?

- a. A method with a detailed implementation.
- b. A method with a detailed implementation but cannot be overridden.
- c. A method with no body which needs to be overridden in a derived class to be useful.
- d. A method that can be accessed from anywhere.

18. Which access modifier restricts access to a class member to within the same class only?

- a. public

- b. private
 - c. protected
 - d. default
19. Which of the following methods are required to ensure visibility of frames?
- i. setBounds(int x-coordinate, int y-coordinate, int width, int height)
 - ii. setSize(int width, int height)
 - iii. setLayout(LayoutManager lm)
 - iv. setVisible(true)
- a. i and iv only
 - b. ii and iv only
 - c. i, iii, and iv only
 - d. ii, iii and iv only
20. Which of the following is not the principle of software modeling?
- a. Using a single model is sufficient.
 - b. Choosing the right model is important based on project scope.
 - c. Each software model has a different level of precisions.
 - d. The best models are connected to real world situation.

SECTION B: DESCRIPTIVE QUESTIONS (25 MARKS)
ANSWER ALL QUESTION

QUESTION 1.

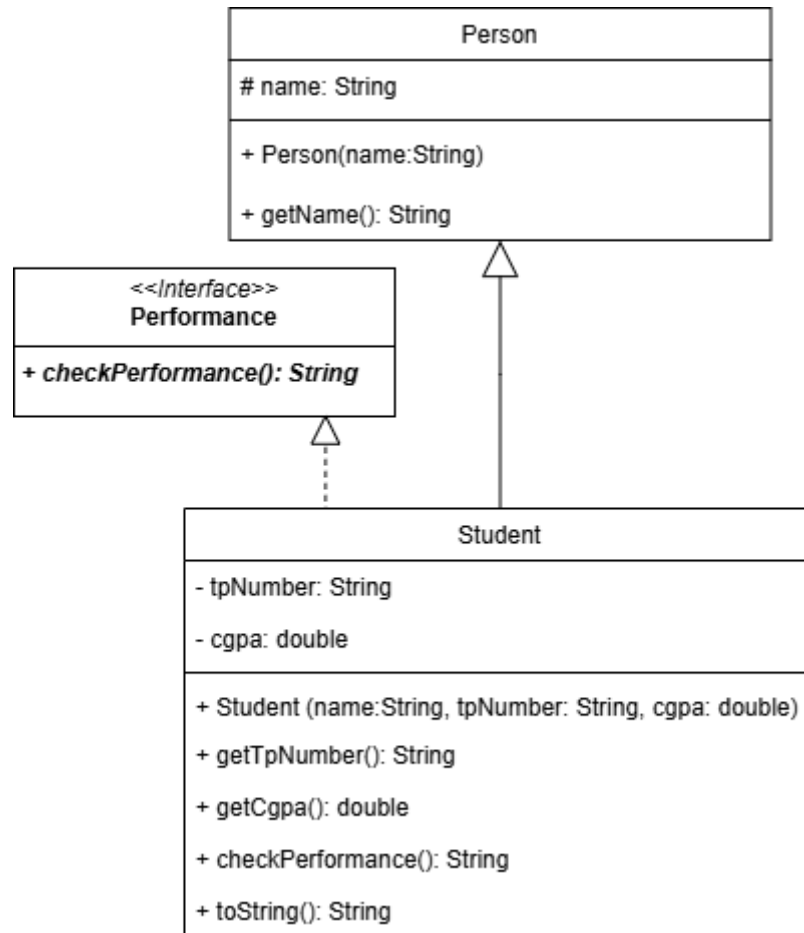


Figure 1.

Figure 1.0 presents a UML class diagram that shows the relationship between a **Person** class, a **Student** subclass, and the **Performance** interface. You are required to write Java code according to Figure 1.0. The '**checkPerformance()**' method will check and return the student's performance status based on the CGPA value. The table 1 shows the performance status corresponding to the CGPA value.

Table 1

CGPA	Performance Status
More 3.5	Excellent
Between 3.0 and 3.5	Good
Between 2.5 and 2.9	Satisfactory
Below 2.5	Need Improvement

(10 marks)

QUESTION 2.

Develop a full Java code based on the following requirements:

Write an abstract class named `Product` that have essential details about a product, including the following attributes:

- **name** (a string representing the name of the product).
- **basePrice** (a floating-point value representing the product's base price).
- **id** (a string uniquely identifying the product).
- **category** (a string indicating the type or category of the product which either `Electronics` or `Food`).

The `Product` class should include a constructor to initialize these attributes with appropriate values and provide getter methods to retrieve their values. Additionally, declare an abstract method named `displayDetails()`, which will be used to output the complete information.

Next, write a class named `Electronics` which inherits from the `Product` class. This class should have two attributes:

- **warranty** (an integer value representing the duration of the warranty in years),
- **brand** (a string indicating the product's brand).

The `Electronics` class should have a constructor to initialize its attributes and include appropriate accessor methods (getters).

Then, write a class named `Food` which inherits from the `Product` class. This class should have two attributes:

- **expirationDate** (a string representing the expiration date of the product).
- **calories** (an integer value representing the food calories given).

The `Food` class should include a constructor to initialize all its attributes and should include appropriate accessor methods (getters).

Finally, create a `ProductTest` class that tests both `Electronic` and `Food` products by creating instances of each class display the details correctly as demonstrated below.

Sample Output

```
Electronic Product Details:  
Name: Laptop  
Base Price: RM 3000.0  
ID: PROD001  
Category: Electronics  
Brand: Dell  
Warranty: 2 years  
Price After Tax: RM 3240.0  
  
Food Product Details:  
Name: Maggi Curry Cup  
Base Price: RM 5.0  
ID: PROD002  
Category: Food  
Expiration Date: 31/01/2026  
Calories: 1200  
Price After Tax: RM 5.4
```

(15 marks)